

A Review on AI-Driven Energy Consumption Forecasting for Smart Buildings

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Received: 18 October 2025 | **Revised:** 16 February 2026 | **Accepted:** 23 March 2026

Keywords: artificial intelligence and data analytics | energy consumption | internet | smart cities | smart power grids

ABSTRACT

Smart buildings integrate advanced technologies with user-friendly interfaces to optimise energy consumption while enhancing occupant comfort. Achieving sustainable living in smart homes relies on motion sensors, infrared sensors, and other intelligent systems that automate and regulate energy usage. While these technologies improve user experience, reducing energy consumption remains a critical challenge for stakeholders committed to sustainability. This paper presents different AI methods for energy consumption forecasting. This review also provides a comparative analysis of various AI-driven, deep learning and machine learning approaches, evaluating their effectiveness in forecasting energy demand and promoting sustainability in smart buildings. This review also gives an overview of predictions on the amount of energy smart buildings will use to achieve the goal of making them more energy-efficient. The study also looks at different ways to predict the future for both residential and non-residential buildings.

1 | Introduction

Buildings use more energy than any other kind of structure around the world. Technological improvements have turned many traditional buildings into 'smart' ones. These 'smart' buildings have saved a lot of energy by using energy management systems, which also makes it easier to keep an eye on and control the building's energy resources. One of the most important parts of these energy approaches is being able to predict how much energy a structure will need. Changing how buildings are used, this makes energy consumption more efficient. People and governments are both very interested in energy efficiency since it cuts down on energy use, cuts down on chemicals that harm the ozone layer, and saves a lot of money.

To cut down on greenhouse gas emissions and encourage building methods that are good for the environment, buildings need to use less energy. Using advanced control methods can make a big difference in how much it costs to run a building and how well its assets are organised. Some systems employ disruptive influence predictions to predict how the building will perform in the future and figure out what to do next [1].

Forecasting is a very important technique for proactive energy management because it analyses all available data to make the best guess about the future [2]. People have been talking a lot about how to make the smart grid more energy efficient lately. Smart grids provide demand-side management features, including building

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energy management systems (BEMS), that let end users adjust how they consume energy in many different ways [3]. Building energy management systems (BEMS) were made to keep an eye on and control a building's energy use in a way that cuts down on carbon dioxide emissions [4, 5].

As building energy management systems have gotten better over the last few decades, it has become more and more crucial to be able to predict how much energy a building will use [6, 7].

The effectiveness of BEMS is affected by vulnerability problems that arise from the changing behaviour of the people who live there. To address this security concern, it is suggested that BEMS incorporate forecasts as a module [8]. The prediction of how much energy a building will use is the basis for improving how well it works through low-energy and control measures [9].

An accurate forecast is essential for improving building energy utilisation and securing substantial energy reserve funds, as the prediction directly influences the control systems and the building's potential energy reserves.

Smart buildings use advanced automation and supervisory systems to run and manage a number of important functions, including HVAC, lighting, and systems for protecting life. Interoperable communication protocols make it easier for these smart systems to share data between domains. It is very important to be able to spot occurrences in one subsystem and then take steps to make up for them in other systems that are linked to it. This is done by using real-time analytics engines to look at data from a variety of sensors and get important information, such as the weather and the number of people in a room. So, adaptive learning algorithms are utilised to change control policies online all the time depending on this data. The ultimate goal of this architectural design is to produce an occupant-centred operation that protects health and comfort in terms of temperature and light. At the same time, these technologies lower both energy use and the greenhouse gas emissions that come with it [10].

The requirement to improve the energy efficiency of structures. This shows that the building industry has a big impact on the environment, accounting for about 40% of the world's energy use and 36% of CO₂ emissions [11].

Adding machine learning to building management systems has been a tremendous and game-changing step in dealing with these big energy problems. Machine learning algorithms can handle a lot of data from different building systems, such as HVAC, lighting, and occupancy sensors. Because of this processing, they can detect complicated patterns, accurately guess what will happen in the future, and change operating parameters in real time. The shift from rule-based control systems to data-driven, adaptive management solutions that can smartly change to changing situations and how people act is a big step forward [12].

Machine learning-driven systems, on the other hand, are always getting better and changing. Traditional building automation systems can't adapt to new scenarios or learn from past mistakes. This ability to keep learning makes it easier to employ renewable energy sources, use energy efficiently based on real-time occupancy data, and generate better predictions about when equipment may break.

The earlier studies have been based on simple statistical models, such as ARIMA, ARMA, and exponential smoothing. These models provide good computational efficiency, interpretability, and short-term predictive power; however, they cannot capture non-linear correlation and drastic change patterns in energy usage. Hybrid models such as GM-ARIMA and ARIMA-SVR were developed with the goal to benefit from both statistical and non-linear approaches, enhance forecasting precision, and overcome some limitations partly. With the advent of AI, machine learning, and deep learning techniques, researchers were able to capture complex, dynamic, nonlinear structures in energy consumption data. Because of these models' ability to learn from past data, they can achieve better accuracy and generalisation. Hybrid methods, which combine the AI-based and traditional approaches, attempt to leverage the complementary strengths of each method, though their effectiveness is often constrained by high computational costs and potential overfitting.

Outdoor weather conditions are consistently identified as a predominant predictor of energy use in all types of models (conventional, AI, and hybrid), followed by indoor environmental factors and building features. Including such variables in (hybrid or AI-based) frameworks could help enhance model accuracy and generalisability. This review suggests several areas for further investigation:

The main findings of this study are:

- A review of the literature to evaluate the advantages and disadvantages of previous research on energy consumption forecasts in smart buildings.
- Develop and test more advanced deep learning and explainable AI algorithms to further improve predictive performance.
- Incorporate building characteristics and weather variables to improve prediction performance.
- A comprehensive examination of the many AI methods used to predict the energy consumption of buildings.
- A comprehensive examination of the ML methods used to predict the energy consumption of buildings.
- Evaluating the current circumstances and prospective obstacles to accurately predict a building's energy use.
- Identification of some real-time case studies.
- Identification of the current state and future challenges for the forecasting of building energy consumption.

Section 2 of this report explains how the review process works. The energy consumption projections for buildings are based on the methodology, goals, and other indicators described in Section 3. Different forecasting methods have discussed in Section 4. In Section 5, AI, ML, and deep learning methods with advantages are discussed, which are used in smart buildings. In Section 6 stages of some ML methods that are implemented in smart buildings are discussed. In Section 7, case studies are discussed, while in Section 8, challenges and solutions are discussed (Table 1).

2 | Literature Review

The idea of smart buildings is becoming more popular, which makes it easier to find a list of them. This list has traditional homes, sensitive homes, and smart buildings. The author of the study [13] says that construction projects have mostly been

TABLE 1 | List of abbreviations.

List of abbreviations	
EC	Energy consumption
CNN	Convolutional neural network
CV	Coefficient of variation
AEP	American electric power
SMAPE	Symmetrical mean absolute percentage error
NRMSE	Normalised mean square error
DT	Decision tree
GBRT	Gradient boosted regression trees
SCOA	Sine-cosine optimisation algorithm
DRNN	Deep recurrent neural network
SAE	Stacked autoencoder
SARIMA	Seasonal autoregressive integrated moving average
SI	Swarm intelligence
SLTF	Short-term load forecasting
LTLF	Long-term load forecasting
SRM	Structural risk minimisation
TNB	Tenaga nasional behrad
HVAC	Heating, ventilation, and air conditioning
ARIMA	Autoregressive integrated moving average.
VMD	Variational model decomposition
AR	Auto regressive
RF	Random forest
ARX	AR model with exogenous variables
MA	Moving average
VAR	Vector autoregression
VARX	VAR model with exogenous variables
PSO	Practical swarm optimisation
HSBUFC	Hybrid stacked bi-directional uni-directional fully connected
SVR	Support vector regression
LF	Levy flights
ELM	Extreme learning machine
RICNN	Recurrent inception CNN
EEC	Electrical energy consumption
MTLF	Medium-term load forecasting
SVM	Support vector machine
EEC	Electric energy consumption
BDGRU	Bidirectional gated recurrent unit
Bi-LSTM	Bidirectional long short-term memory
EMD	Empirical mode decomposition
MFMOA-NMM	Modified fish migration optimisation algorithm-based numerical moment matching

focused on improving overall performance, lowering energy use, and making it easier for people to interact with each other. Smart homes help owners and renters reach their goals by making it easier to manage resources like space, energy, and the environment [14]. More and more people are interested in smart homes. The author of the study [15] says that the construction industry's ecosystem works better and more efficiently when it adapts to

changes in society and technology that are caused by the needs of people and businesses. Artificial intelligence models are based on biological principles, according to a study [16]. These models are based on how nature learns, adapts, and changes by interacting with its environment. Smart buildings are becoming more popular, and one reason may be that they use creative and environmentally friendly design principles. Additionally, new

technologies are making space, energy, comfort, health, and operational management better for users [17]. This principle is also applicable to operational activities. The explosive growth of coworking spaces in recent years has redefined traditional corporate structures and work environments, as well as interpersonal dynamics.

People, society, management, and businesses have all been criticised for giving people more freedom and power. The author of the study [18] asserts that the notion of construction has transitioned from cleverness to intelligence. This is due to the progress of modern information technologies. Artificial intelligence allows buildings to learn, make decisions, and update themselves without human help. Thus, architectural intelligence is clearly shown in how buildings are run and how they work. The main ideas behind energy-efficient building practices are using and combining smart technologies and setting up different smart information systems [19]. Artificially intelligent buildings are a new idea that is being used in modern construction projects. The development of Artificially Intelligent Buildings (AIB) has been made easier by new technologies like AI algorithms and related systems that help manage energy use in the best way possible.

Many new technologies are always changing how society uses artificial intelligence. It is worth the time and effort to look into the possibilities and risks of artificial intelligence systems because they can be very helpful [20]. People are more afraid and sceptical of AI systems because they are worried about losing control, but this is still true. The main goal of improving artificial intelligence is to make people smarter and better at what they do. The author of the study [21] drives all new ideas and research in the field of smart buildings. There are many goals for building development, such as saving water and energy, making buildings last longer, making people more comfortable, and making them happier. Smart buildings need to be more flexible than they need to be reactive. The study [22] shows that a unified system is being built that includes business operations, control, materials, and construction. These systems might change over time. The intelligent building may change over time by collecting and analysing new data from many different places. The author of the study [23] says that a building's intelligence is made up of its long-term durability, design, safety, comfort, and security. Combining modern infrastructure with advanced data processing, artificial intelligence, and control systems leads to smart buildings [24].

The objective of this work [25] is to create and improve a Multilayer Perceptron Regressor (MLPRegressor) model to predict the heating and cooling loads of a building. The objective of this study is to formulate and advance a model. The model that has been suggested is based on the Waterwheel Plant Algorithm, which optimises hyperparameters. The model was based on a set of data that described how the structure used energy. Several important characteristics are considered input factors. People think these variables are very important. Some examples of these are the overall height, the roof area, the wall area, the orientation, the distribution of the glazing material throughout the structure, and the relative compactness.

This study [26] predicts building energy use and indoor climate using CNNs and LOA. The research gives architects and

engineers real-time building performance optimisation tools using sub-hourly data to meet escalating energy efficiency and interior environment needs. This research presents a new way for improving prediction model accuracy and efficiency using CNNs' spatial and temporal data processing and LOA's systematic feature selection.

This literature review [27] aims to examine the revolutionary impact of Internet of Things (IoT) technology on smart building management systems, focusing specifically on enhancements in operational efficiency, occupant safety, and comfort. Technological advancements streamline and automate critical activities, including energy management, security, and environmental regulation, while the proliferation of Internet of Things applications in constructed environments persists. Despite the advantages, including the integration of the Internet of Things (IoT) in smart buildings, various challenges persist. This encompasses system compatibility, infrastructure scalability, and issues of data security and privacy within a connected environment.

The paper [28] says that blockchain technology might be able to fix the biggest problems with IoT sensor systems in smart buildings. There are many benefits to using blockchain technology, such as better data security, interoperability, scalability, and faster processing in real time. Blockchain is a decentralised, open, and unchangeable system that keeps data collected by Internet of Things (IoT) sensors safe from snooping and tampering.

This study [29] proposes a model of research-based lobbying led by Civil Society Organizations (CSOs) and supported by tactics focused on defending human rights. It pushed for the ignored right of people living in thermally-insulated houses in Jordan to have walls that don't get too wet, mouldy, or uncomfortable, and that don't cost a lot of money to heat.

The suggested framework [30] demonstrated exceptional prediction performance, evidenced by a Mean Absolute Error (MAE) of 0.237 kWh, a Root Mean Square Error (RMSE) of 0.298 kWh, and an R-squared (R^2) value of 0.978, indicating a 97.8% explanation of data variability. The classification metrics showed that the model was even more robust. They had an accuracy of 97.7%, a precision of 97.8%, a recall of 97.6%, and an F1 Score of 97.7%. A comparative investigation of standard models, such as Linear Regression, Random Forest, Support Vector Machines, Long Short-Term Memory, and XGBoost, indicated that the proposed method exhibited enhanced accuracy and real-time adaptability.

The study [31] main goals are to examine the historical difficulties with building management systems and evaluate the possible benefits of integrating blockchain and artificial intelligence (AI) into current initiatives for building sustainability, security, and efficiency. The narrative review method looked at a wide range of case studies, books, and procedures. Notable innovations include AI's ability to improve current functional systems, such as lighting, HVAC, and predictive maintenance. As an alternative, blockchain technology offers decentralisation, immutability, and transparent data management.

This solution [32] leverages natural human interaction and autonomous building operations management by merging

blockchain governance, digital twins, and AI agents and assistants that can handle smart building appliances without human supervision. It also leverages blockchain-based protocols to build an infrastructure that is managed by everyone and works on its own. The prototype's ability to support autonomous, secure, and community-governed building systems was tested in both simulated and real-world situations. This dissertation aims to contribute to the growing body of research at the intersection of smart infrastructure, generative AI-driven automation, blockchain governance, and human-building interaction.

This review [33] provides a comprehensive analysis of 104 works and a synthesis of the current literature to evaluate the challenges posed by Internet of Things sensors and the application of Blockchain-based technology to mitigate these issues. This analysis seeks to address the absence of a building-specific sensor-layer synthesis that systematically links Internet of Things (IoT) sensor issues in smart buildings to tangible blockchain architecture solutions. We look into the market and make a structured map of the challenge mechanisms that deal with energy, maintenance, localisation, privacy and security, scalability, interoperability, real-time processing, restricted artificial intelligence integration, and false positives. This makes it easy to construct management systems that are safe and work well.

An effective interior environment in a smart building can be achieved through the integration of intelligent design, management, amenities, and their synergistic functionality. Researchers primarily concentrate on the application of ICT, interpersonal interactions, and operational efficiency. Researchers look at new developments in AI, like smart buildings, in CiteSpace. The next sections go into more detail about the different ways that smart building technologies can be used to run buildings. The author of the study [34] contends that the Internet of Things (IoT) can be employed to change the way civil engineering and infrastructure are usually done. Technological progress has made it possible for digital twins, smart cities, smart environments, and new ways to design buildings to become a reality. It's hard to talk about data management and IT security. More and more people are interested in advanced topics like data analytics in smart buildings and cities, as well as designing smart environments with the Internet of Things (IoT). This will change how people think about architecture and environmental engineering that uses AI.

There has been little talk about how self-driving AI agents and training environments could make learning more efficient in complicated and changing operational settings. The ability of buildings to learn at a systemic level sheds light on autonomous machine learning programs that independently control building energy. The author of the study [35] says that two ways to improve are to use digital twins as training tools and to make systems that can quickly adapt to changes. People will be able to enjoy more comfort and convenience in smart buildings in the future. Smart building appliances and devices create and store huge amounts of streaming data that need to be turned into useful information so that decisions can be made quickly.

It would not be possible to offer these smart services without a lot of data analysis. The author of the study [36] gives a thorough look at the smart buildings industry, looking at the main trends and problems that come up when trying to improve smart building

services. They focus on how important the main data analytics methods are.

People all over the world are coming up with and using strategies based on artificial intelligence (AI). The author of the study [37] contends that LDFT is capable of handling more complex input and output relationships. This study offers an exhaustive examination of the literature about artificial intelligence-based short-term load forecasting methodologies. The main goal of this study is to look at current research on AI-based load prediction models and see how well they work by testing and analysing them. The Fire Monitoring and Warning System (FMWS) uses fuzzy logic to let the Fire Management System know about fires that could be dangerous. This study investigates a Fuzzy Logic Fire Monitoring and Warning System that communicates an alarm notification through a Global System for Mobile Communication. The author of the study [38] created the system using small, cheap sensors to make sure that the responses were always the same.

Python is used to create and improve a model that predicts how much energy a device will use based on factors like humidity, air speed, radiant temperature, and thermal comfort indicators. As [39] noted, there is a new trend in IT development that has made it common to use smart techniques to guess how much energy a building will use. In a related study [40], examines the capacity of cloud computing to provide energy-related services that improve the effectiveness of smart building energy management systems (EMS). Cloud computing can improve smart building energy management systems by making it easier to store software models that are needed to use different control, monitoring, and optimisation methods to cut down on energy use in buildings. We look at the pros and cons of using cloud computing to make smart buildings more energy efficient. The author of [41] asserts that the design, architecture and experimental validation must integrate cloud-based AI-driven monitoring platforms and the Internet of Things (IoT). The Cloud Cyber layer is the basis for the proposed five-tiered platform for Industry 5.0. The author of the study [42] talks about fuzzy logic, smart building architecture, and the Open Fog Standard. This study is undertaken due to the urgent need to reduce energy consumption in both public and private buildings while improving comfort, safety, and the internal environment. To reach this goal, advanced computer methods will be used. This study describes an open, layered architecture for smart buildings that meets the IEEE 1934–2018 standard. The Open Fog Reference Architecture is what this is built on. The proposed design, which uses fuzzy logic to improve indoor air quality, was tested at the Vitoria-Gasteiz Faculty of Engineering. The test results showed that the design worked and could be made bigger.

The author of the study [43] talked about AI-powered buildings that are very accurate and the Monthly Energy Benchmarking Approach (MEBA). At first, MEBA could only use monthly data to figure out how much energy was used each year. Second, it can figure out monthly loads by looking at data from the whole year. The study gathered monthly data on the use of electricity and natural gas from two American cities. The first case says that the author uses a Gaussian Mixture Model (GMM) to split the dataset into two main groups. Self-organising Map (SOM) models are used to divide the two sets into five more groups

based on how much energy they use. In the second scenario, an extra step is needed to find the subcluster labels using advanced Light Gradient Boosting Machine (LGBM) classifications. The five subclusters together make predictions that are more than 95% accurate. In both cases, the last step is to use LGBM regressions to guess how much electricity and natural gas will be used each month. The MEBA model is better at making predictions than the most advanced XGBoost model, with R^2 values between 0.50 and 0.73 and RMSEs between 0.15 and 2.35. The most important things about each sub-cluster's unique energy use patterns are its electrical loads, the year it was built, and its Energy Use Intensities (EUIs). Using AI in the design and building process has a lot of benefits. Despite some difficulties, artificial intelligence (AI) has become a powerful force for modernising the construction industry. AI can help the construction industry become more sustainable by making its processes and practices more efficient.

The author of the study [44] suggested the use of Explainable Artificial Intelligence (XAI) to create a complex, safe, and clear energy management system that uses federated learning. Federated Learning decentralises model training to improve data privacy and security, while Explainable Artificial Intelligence

clarifies AI decisions for users to build trust. Federated Learning (FL) and Explainable Artificial Intelligence (XAI) work together to create a safe, clear, and understandable way to control how much energy smart buildings use. This research introduces a federated learning-based explainable artificial intelligence model aimed at bolstering user confidence by enabling expedited, accessible, and efficient decision-making while ensuring data privacy. Simulations demonstrate that the proposed model surpasses the alternative methods assessed. The author of the study [45] suggested evaluating the energy efficiency and comfort levels of windows employing PDLC in an Istanbul office building. Table 2 represents the comparative analysis of deep learning and machine learning models in terms of data requirements, computational complexity, strengths, limitations, and generalisation capability.

A comparative study of the frequently used machine learning and deep learning models for energy consumption forecasting is listed in Table 3. This comparison is made in various aspects, including limitations, strengths, data requirements, computational complexity, and their capacity for generalisation. Conventional models, including SVR and methods based on ensembles, show good interpretability and efficiency when it

TABLE 2 | Comparative analysis of AI-based models.

Model	Strengths	Limitations	Data requirements	Computational complexity	Generalisation capability
SVR/Linear SVR/SVM	Good with high-dimensional data; effective for small to medium datasets; interpretable	Sensitive to parameter tuning; may not scale to very large datasets	Small to medium	Medium	Good
ANN/FCNN/MLP/IGA-ANN	Can model complex non-linear patterns; flexible architecture	Black-box; sensitive to hyperparameters	Medium to large datasets	Medium to high	Moderate
LSTM/LSTM-attention/ConvLSTM/GRU/VMD-LSTM/TA-EDN/Deep LSTM/LSTM-RNN/Pooling-based DRNN	Captures temporal dependencies and sequence patterns; suitable for long-term forecasting	Computationally intensive; requires large datasets; slow training	Large	High	Good
Bayesian Optimised-LSTM	Hyperparameter optimisation improves performance	Computational overhead; complex setup	Medium to large	High	High
CNN-LSTM/RICNN/ConvLSTM with BDGRU	Captures spatial-temporal dependencies; effective for complex sequences	High computational cost; requires large datasets	Large	High	Good
RNN	Good for sequence modeling	Vanishing gradient problem; less effective for long sequences	Medium to large	Medium	Moderate
ELM/Deep ELM/SAE-ELM	Fast training; reasonable generalisation	Limited depth for very complex patterns; sensitive to noise	Medium	Low	Good
Decision trees/random forest/AdaBoost/ensemble methods	Robust; interpretable; handles categorical features well	Can overfit with small datasets	Medium	Low to medium	High

TABLE 3 | Summary of the hybrid method.

Ref.	Contribution	Limitation
[46]	A system that predicts how much power will be used by using statistical and artificial intelligence models, as well as the data entropy-based weighting method.	This model is suitable solely for projecting power consumption over a medium-term horizon.
[47]	A new way to estimate electrical demand that uses a cutting-edge algorithm for shark odour progression and a unified augmented Elman neural network.	The model should only be used to make short-term predictions.
[48]	To anticipate the electrical needs of a non-residential structure, a hybrid method that uses both K-means and autoregressive integrated moving average (ARIMA) is applied.	The purpose of this method was to cut down on the amount of electricity that non-residential buildings used.
[49]	A method for combining transfer learning and long-short-term memory (LSTM) to figure out how an intelligent building uses different kinds of electrical energy.	The procedure was only tested on homes.
[50]	A convolutional neural network and a larger short-term memory part work together to anticipate how much energy buildings will need in the future, both homes and businesses. the encoder.	There were two datasets: One used 1 m to record energy use, and the other used 3 m.
[51]	The two-hybrid forecasting approach for univariate time series is a good way to find out the cooling capacity and other systems.	Finding the temporal lag makes the model's approach a little more complicated.
[52]	A method for predicting energy use that leverages load data and physical knowledge to deal with a range of possible outcomes utilising GP regression and heuristics.	That can get a prediction that is one to 5 days ahead of time in as little as 10 min.

comes to small- or medium-sized datasets; meanwhile, methods using neural networks arrive at higher performance due to the increased flexibility in learning non-linear relationships. Temporal and spatio-temporal long-term dependencies can be well represented by the recurrent and deep learning architectures, for example, LSTM, CNN-LSTM networks, but they depend on bigger volumes of data and computational resources. Consequently, Bayesian optimisation has been proposed to automatically optimise the hyperparameters for improved performance, but with added computational complexity. This comparative study highlights the accuracy, computational complexity, and scalability trade-offs, which in turn motivate us to use state-of-the-art deep learning models for enhanced forecasting.

3 | Building Energy Consumption Forecasting

One purpose is to check and measure how much energy-saving programs [53] have worked. The purpose of this objective is to evaluate the effectiveness of these measures in reducing energy consumption after the formulation and implementation of the associated recommendations.

- The guidelines yield the anticipated outcomes.
- Presume energy requirements under standard conditions, aiming to forecast events that may disrupt the usual energy consumption of the systems (e.g., a sudden temperature change that impacts occupant comfort)
- Specify variations in energy consumption during particular events, intending to measure the extent of change in energy usage following an event that alters these predictions to plan how to distribute and arrange products to fulfil demand [54].

In order to optimise intelligent building operations, manage energy resources, find system problems, and make buildings work better, it is important to have accurate estimates of energy use [55, 56]. This part of the control system often helps it figure out the appropriate control settings to reach energy or cost investment goals. The accuracy of estimations of energy use has a big effect on how energy and cost investment funds are distributed and how general regulations are put into place [57]. Figure 1 shows the many goals that building forecasting programs try to reach.

The building owners make more money because of better predictions. An unexpected gap makes the assessment harder, which leads to wasted money and poor energy management. Because of this, it is very important to improve estimates of energy demand [58].

The main goal of building forecasting is to produce forecasts about how many people will be living in a building, what the weather will be, and how much energy it will use in the future. This will help improve the design and operation of structures. Using accurate predictions from building managers and experts, it is possible to improve the building's total energy efficiency, cut down on operational costs, and save energy. It lets you plan for HVAC systems by guessing how much demand there will be in the future and changing the way you operate them.

One of the most important ideas in sustainable building management is to cut down on carbon pollution and use renewable energy sources. Forecasting can help stop wasting energy and overloading systems. It can also help in managing workloads and lowering peak demand. It also makes people more comfortable by anticipating the temperature and air quality in buildings, which

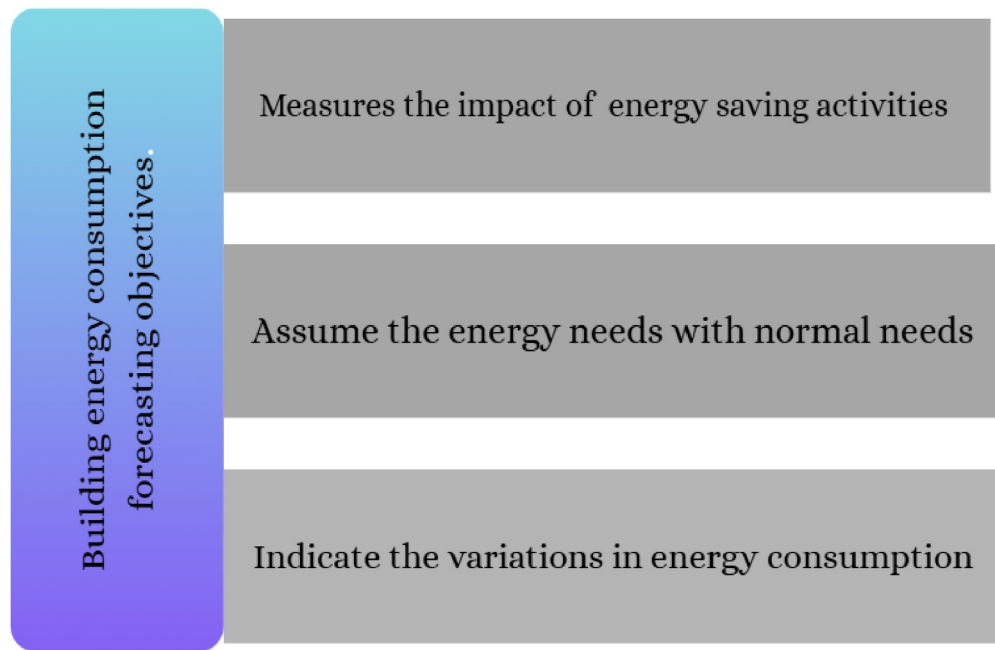


FIGURE 1 | Building energy consumption forecasting objectives.

makes it easy to make changes quickly to keep living and working areas in the best possible condition.

Forecasting in buildings can help with planning maintenance and figuring out when equipment might break down and when preventative maintenance should be done. It is also very important for smart building systems and energy management tools because it lets you make decisions based on data. Forecasting in buildings leads to proactive and well-informed management strategies that make buildings work better, are better for the environment, and save money.

4 | Forecasting Methods

4.1 | Physical Method

Forecasting systems can accurately predict energy use by linking energy to the system that uses it [59]. This helps reduce waste caused by having too much or too little of something. Several ways have been suggested in recent years to deal with this information.

According to forecasts [60]. Smart meters are widely used in the construction industry to keep track of and report on energy use. These make it easier to predict the future in the energy sector and give new ways to use [61]. There is a solid reason to predict how much energy a building will use when it comes to managing and improving energy efficiency. Accurate energy forecasting systems [62] have several effects on planning and energy optimisation. In the last 20 years, a lot of building energy forecasting systems have been made, each with its own point of view [63].

There are three main parts to building energy modelling and forecasting: physical, data-driven, and hybrid methods [64, 65]. Thermodynamic methods can be used to figure out how much energy a building uses, which is often nicknamed a ‘white-box’ because its systems are easy to see [66]. The ‘black-box’ method,

which is based on statistics, forecasts how much energy a building would use without taking into account its physical features. It uses machine learning and statistical time series analysis to figure out and predict how much energy the building will need [67]. A grey-box methodology combines methods based on data and methods based on the physical world. It was first released in the 1990s to make HVAC control systems work better [68].

Physical approaches let you slowly evaluate the cycle by modelling and monitoring it in a controlled setting where all the parameters are known [69]. Grand predictions fail because they can’t deal with the intricate changes that happen when people do things. These models are not utilised very often because they have trouble making accurate load predictions [70]. It’s hard to get all the information you need to come up with a good physical strategy because there are so many input elements.

Also, the data is not accurate or reliable, which means that the model outputs do not match reality [71]. Studies on demand forecast models for various energy sources have concentrated on physical methodologies, including the construction of lift energy consumption, cooling electrical consumption, energy demand, load forecasting, electric and heat load, and heating consumption. Table 4 shows the previous studies on the physical method.

4.2 | Data-Driven Method

Data-driven methods usually work better than other methods when it comes to being easy to regulate, having cheap planning expenses, and being easy to use. The high number of dimensions in the data makes it hard to guess which method will work best for data-driven approaches [81]. Figure 2 shows the forecasting methods.

A lot of research has been done on data-driven methods in relation to energy use [82, 83], indoor thermal comfort, electricity use,

TABLE 4 | Previous research on the physical method.

Ref.	Contribution	Limitation
[72]	There are numerous methods for evaluating the annual energy consumption of commercial buildings.	Ultimately, the techniques that are proposed will prove to be ineffective for long term.
[73]	An accurate forecasting model for lift efficiency that is based on SARIMA	In the course of the analysis, variables that were associated with schedules, energy levels, and social challenges were not taken into consideration.
[74]	A method for figuring out a building's national cooling burden and then looking at how much power it uses in two different situations.	The other variables did not have any effect on the year-average temperature.
[75]	System identification is one way to figure out how much energy it takes to cool an entire building.	The study focused solely on one demand response action for precooling due to its analytical objective.
[76]	A two-part technique is used to figure out how much energy urban buildings need.	We didn't think about how people might react when we came up with the notion. We thought that all buildings, no matter what type they were, would use the same amount of energy.
[77]	Using flight schedule data, regression analysis is performed to predict how much electricity airport facilities would need.	The analysis assumed that every seat on a predefined skyline was full to figure out how many people were on board.
[78]	A way to predict and show how much energy homes will use through 2040.	We don't know enough about how people use their homes.
[79]	A strategy for managing energy in a commercial facility that can predict short-term energy needs.	Researchers looked into how easy and cheap it is to use different load assessment methods. The research did not examine the precision of the forecasts.
[80]	Methods for factoring in changes to building inventories when figuring out how much power an area or country would need over the long term.	The characteristics of industrial structures and the cargo of electric vehicles were not analysed.

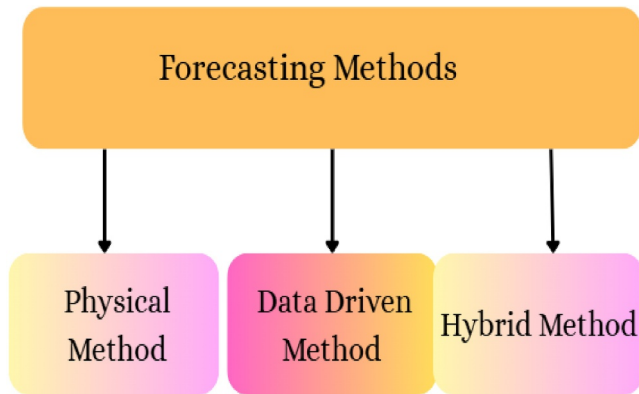


FIGURE 2 | Energy Forecasting methods in smart buildings.

building photovoltaic output [76], cooling load, heating and cooling needs, occupancy and energy use [83], clustering of energy use and peak load demand [84]. Table 5 gives a summary of the data-driven method.

4.3 | Hybrid Methods

Hybrid methods necessitate techniques capable of identifying and representing the schedule, including variations in plug loads, lighting, and occupant density [92]. The hybrid approach looks at how to integrate the best parts of two worlds: boundary estimation and a measurable framework, and physical understanding and model structure, when it comes to structures that already exist and have monitoring data [93]. The boundary advancement technique and the master information needed to improve the model can make utilising a hybrid approach take a long time to

process [94]. Studies on hybrid approaches have looked at a number of things, such as energy use [95, 96], load demand [97], heating and cooling load, electricity use, thermal load and thermal response. Table 3 highlights the hybrid method.

5 | Role of AI and Deep Learning Techniques in Smart Building

Artificial intelligence (AI) and deep learning (DL) are important for smart building resource management because they can help us get measurable results in terms of efficiency and sustainability. There are a lot of different types of applications in use right now, and some of them are very precise. Using occupancy detection and energy-consumption estimates makes it possible to fully tailor HVAC control.

One of the most important things that AI must do is make predictions. For big regression and classification tasks, traditional machine learning algorithms like support vector machines (SVM), k-nearest neighbours (KNN), and decision trees still function effectively. On the other hand, ensemble learners like gradient boosting frequently get better accuracy levels [36, 98]. Recurrent neural networks (RNNs), especially LSTM units, are very good at analysing time series data, like energy consumption computations [99, 100]. But convolutional neural networks are very excellent at finding patterns in camera feeds and several types of sensor arrays.

Reinforcement learning (RL) and deep reinforcement learning (DRL) are becoming more and more popular because they can make control techniques that adapt to unexpected and changing situations in buildings [101]. When employing algorithms like Q-

TABLE 5 | Summary of the data-driven method.

Ref.	Contribution	Limitation
[82]	Using weighted multiple regression, an SVR model estimates how much energy non-residential buildings will use each day and every half hour.	It could be able to connect the difference between real quality and other metrics. As a result, the model became more complicated, which made it take longer to make.
[85]	A method that employs artificial neural networks (ANNs) to predict how much power a group of buildings or a single building will need.	The model was originally meant to be used to make predictions that would only last a limited time.
[86]	A forecasting model can help you estimate how much electricity, gas, and cooling a building will require.	Overall, the forecasting system did a good job, but it did not accurately estimate the amount of power required for cooling.
[87]	This method uses a Gaussian process (GP) to produce predictions about net demand, solar power generation, and energy use.	The dynamic approach may have had trouble accurately forecasting unexpected peaks in time because it had a limited amount of time to prepare.
[88]	A method that employs an artificial neural network (ANN) and component extraction to predict the power demand profile for a non-residential or commercial building one day in advance.	Once the system was put into action, it showed how much power would be used the next day, broken down into smaller time periods.
[89]	A model that uses an artificial neural network to figure out how much weight a building can hold, taking into account peak loads that aren't always the same and the chance that weather forecasts will be wrong.	The load forecast predicted both conventional and unusual maximum differential loads.
[90]	A method for figuring out how much electricity a healthcare institution will need in the short term based on detailed weather data.	The technique can only be used on buildings that have the same climate and energy profiles.
[91]	It suggests a methodology for generating probabilistic forecasts of solar power production one day ahead, specifically for building energy management system applications.	The process was tested in a certain environment and climate.

learning and deep Q-networks (DQN) to get the optimal HVAC schedules or ways to charge electric vehicles from interaction data [102, 103]. By adding specific information, like physics-based models, to the DRL training loop, it is possible to minimise training time by a factor of 8.8 while keeping thermal and indoor air quality parameters within the comfort range [104]. Using Graph Neural Networks (GNNs) and Deep Reinforcement Learning (DRL), hybrid systems can make HVAC set-point suggestions better.

This is especially true for state spaces with more than one dimension. Deep learning and artificial intelligence are also the basis for predictive maintenance programs for HVAC systems and other building equipment [97]. By finding problems early, continuous sensor-data analysis makes it easier to convert from maintenance based on a calendar to maintenance based on conditions. The Maintenance 4.0 (M4.0) paradigm integrates big data analytics, the Internet of Things, and artificial intelligence to make technical maintenance more successful [105]. The combination of artificial intelligence (AI) and digital twin (DT) technology, which is powered by real-time data from the Internet of Things, makes it easier to keep an eye on things. This allows for quick predictions of events and spotting of strange things [106, 107]. For instance, SVM and LSTM models that are part of a hospital decision tree may change the ventilation based on how many people are expected to walk through.

Even with these changes, there are still numerous problems that need to be rectified. As people become more worried about the

safety of personally identifiable information, new methods like federated learning and differential privacy have become more widespread, often used with edge architectures [99, 108]. Due to issues with data quality and the vast amount of streaming data, it is necessary to set up tight systems for cleaning, curating, and processing data in real time [108]. Also, there isn't enough interoperability; standardized protocols are still needed for different BMS platforms and IoT devices to work together [109, 110]. Ultimately, challenges of model interpretability and user credibility remain unresolved. Nonetheless, investigations on scalable and resource-efficient XAI solutions capable of operating on constrained edge devices remain in their nascent phase [20–45, 53–85]. Table 6 shows the AI and Deep Learning Techniques in Smart Buildings.

Though prior assessments have only examined a small portion of AI's potential applications in smart buildings, it is evident that AI offers numerous advantages. To determine comfort and demand, several researchers employed more conventional machine learning pipelines, although they did not provide any numerical comparisons. Consider how embodied energy or other kinds of algorithms are impacted by AI processes [116, 121]. Data processing and sensor integration were the focus of earlier research, but algorithmic and edge deployment context differences were not taken into account [122, 123]. Deep learning reviews arranged models according to their architecture, but they did not consider how easy it is to grasp them or tactics that use transformers or LLMs [110]. However, the carbon footprint and scalability of massive model trade-offs were not examined in evaluations that concentrated on

TABLE 6 | AI and deep learning techniques in smart buildings.

Ref.	Year	Method	Key findings
[111]	2022	Deep learning	This study talks about current problems and how researchers might leverage deep learning technology to develop FMM-specific deep learning models. The work emphasises the necessity of developing publicly accessible FM datasets to enable deep learning systems to effectively facilitate predictive maintenance. Deep learning technology should be employed for predictive maintenance of thermal-storage air-conditioning (TS-AC) in HVAC systems. This would make construction more cost-effective and help protect the environment.
[112]	2022	Deep learning (CNN, RNN)	In this research, this study investigates the applications of CNN and RNN/LSTM that are used in smart homes, and we compare how these two types of neural networks were developed and evaluated. The findings of this study give crucial data for all researchers who are interested in applying deep learning to smart homes. It also identifies the primary trends and can assist in guiding design and evaluation decisions for specific smart home services.
[35]	2022	Deep reinforcement learning (DRL), supervised/unsupervised ML	This review article discusses autonomous machine learning applications that make decisions on their own for managing building energy. It also talks about how structures can learn from a system-level point of view. The study concludes that the adaptability of buildings to unforeseen changes can be enhanced at the system level through the use of digital twins as training environments and the incorporation of learning processes activated by artificial intelligence.
[113]	2022	Deep learning, transfer learning	This article primarily examines deep learning (DL) and transfer learning (TL) methodologies for occupancy detection, rather than other alternatives. The objective of this study is to evaluate several occupancy detection methodologies to develop an effective system for processing sensor data while simultaneously delivering precise occupancy information. The study also compared the most common occupancy detection algorithms to see which one was the most efficient in terms of training time and test accuracy. To restrict or reduce privacy problems that can come up during occupancy detection, a number of suggestions were offered regarding the way to employ new technologies, including edge devices, federated learning, and the blockchain-based Internet of things.
[114]	2019	Smart building technologies	This research paper looks at the best current approaches for improving intelligent features, managing parameters, and Internet of things (IoT) infrastructure, which are all necessary for smart buildings. The major goal is to sense and control the Internet of things infrastructure. This lets cloud clients use a virtual sensing infrastructure when they use communication protocols. Sensors for safety, general administration, network architecture, privacy and security, and health services are some of the smart parts that are often found in smart buildings. These are only a few examples of the smart features that are usually there.
[115]	2021	Artificial intelligence	This paper seeks to provide a comprehensive evaluation of contemporary research regarding the implementation of artificial intelligence (AI) technologies in smart buildings, utilising the frameworks of building management systems (BMS) and demand response programs (DRPs). An evaluation framework is provided and used to analyse recent research in this area and in the core AI fields of energy, comfort, design, and maintenance. It also talks about the theories and usage of AI-based modeling approaches that are often used to anticipate how much energy a building will require. The scholarly study ends with a look at the problems that still need to be solved and the possible paths for future research on how to employ AI in smart buildings.

(Continues)

TABLE 6 | (Continued)

Ref.	Year	Method	Key findings
[116]	2025	Deep learning	The deployment of productive renewable energy systems for smart cities, smart territories, and other environmentally friendly technologies is a challenging endeavour. One of the primary approaches to accomplish this objective is through the integration of the Internet of things and deep learning. The use of DL already makes it possible to achieve a prediction accuracy of more than 90%, and in some instances, even 98%. It is expected that these values will increase in proportion to the number of devices that are subject to validation and monitoring. This will ensure that they are completely predictable, not only in terms of the amount of power consumed during normal operation, but also in terms of the various sorts of failures and even alternative scenarios.
[117]	2025	Reinforcement learning (RL)	The study analyzes and summarises the various features available in the field for evaluation purposes, considering building typologies, RES types, BEMS types, baseline control approaches, agent types, value-action networks, reward functions, and reinforcement learning techniques. The report talks about the pros and cons of RL in energy management and gives a clear picture of current research trends and possible future directions. This is based on what was said in the paper's evaluation section.
[118]	2025	IoT, explainable AI (XAI)	This study seeks to analyse the influence of the convergence between the Internet of things and artificial intelligence (IoT-XAI) on the decision-making processes, elucidations, and forecasts of customer perceived value (CPV) within small and medium-sized enterprises (SMEs), utilising theories from both CPV and IoT-XAI. This study investigates the influence of customer interaction with Internet of things (IoT) devices on the prediction of consumer purchase volume (CPV) using deep learning (DL) models and the role of explainable artificial intelligence (XAI) in decision-making. We were able to create a theoretical framework for studying the links between IoT and XAI convergence, as well as CPV prediction, explanation, and decision-making in the direction of personalised marketing, by doing a literature review on customer-Internet of things interaction, Internet of things physical objects, Internet of things data analysis, deep learning models, the Internet of things, and customer purchase value.
[119]	2024	IoT sensors, ML, DL	This paper aims to provide a thorough overview of the latest methods and devices for occupancy detection, encompassing both traditional algorithms and machine learning techniques, as well as their respective advantages and disadvantages. The research analyzes and assesses essential technologies for smart buildings, encompassing sensors, algorithms, and related technologies. Furthermore, the survey provides insights and discussions that can aid academics, practitioners, and stakeholders in the advancement of more effective occupancy detection technology for smart buildings.
[101]	2025	Machine learning, deep learning, RL	This study seeks to deliver a comprehensive analysis of the current trends in machine learning-driven energy systems research, elucidate the challenges linked to these advancements, and propose recommendations and potential future directions. Unlike previous studies that largely look at how machine learning is used in the renewable energy sector (RES), this one looks at how ML-driven methods can be used in a wider range of energy systems, such as energy policy and sustainability. This study seeks to be a complete resource that connects theoretical advances with real-world uses in energy systems by using machine learning-driven innovation.
[97]	2025	Deep reinforcement learning, GNN	This paper introduces an innovative hybrid model that integrates graph neural networks (GNN) with deep reinforcement learning (DRL). The dynamic response learning (DRL) part makes adaptive decision-making possible by letting sensor nodes change in real time depending on network performance data. During this time, the GNN model makes coverage more

(Continues)

TABLE 6 | (Continued)

Ref.	Year	Method	Key findings
[108]	2025	AI, ML (CNN, LSTM), federated learning	effective by improving spatial awareness by understanding how sensor nodes are related to each other. This integration makes the network much more efficient and flexible. The proposed DRL-GNN model can achieve a coverage ratio of up to 96.4%, an energy efficiency of 95.8%, and a low overlap of 5.2%, which is better than what standard methods can do. The paper begins by examining the impact of a worldwide pandemic on developing nations, subsequently elucidating the significance of AI in epidemiological modeling, its role in enabling data-driven decision-making, and its contributions to pandemic prediction, prevention, and response. AI-driven epidemiological models like SIR (susceptible-Infectious-recovered) and SIS (susceptible-Infectious-susceptible) help with all of these tasks: Predicting how diseases will spread, stopping outbreaks, and making sure vaccines are given out in the best way possible. The review also highlights how machine learning (ML) algorithms and predictive analytics can help us figure out how illnesses spread. AI's role in making vaccines and running clinical trials is emphasised, with a focus on making AI-powered surveillance networks.
[120]	2019	Deep reinforcement learning	This paper proposes the utilisation of 'experience replay' and/or 'expert policy guidance' to encode expert information within reinforcement learning frameworks. Also, controlling the smoothness of the neural network to punish erratic behaviour makes the learning process more stable and results in control laws that are easy to grasp. This work created a virtual testbed for building control using python and EnergyPlus, a state-of-the-art building energy model. This will help us look at and evaluate current building control algorithms, figure out what works and what doesn't, and suggest areas for future research.

reinforcement learning [124]. As such, we were unable to provide a thorough assessment of the current status of the field. Therefore, this paper aims to summarise existing knowledge, discuss current applications of AI and deep learning in smart buildings, and identify challenges that remain to be addressed.

6 | Role of Machine Learning Models in Smart Buildings

Machine learning has drastically transformed the way we save energy and regulate smart buildings. Research on how machine learning may be used to predict energy use shows that these technologies are not only extra tools; they are necessary for achieving operational excellence, sustainability, and energy efficiency. Energy forecasting is a key feature of building energy management systems (BEMS). By forecasting future energy needs, the goal is to get the most out of the system, spend the least amount of money, and keep it stable. Machine learning (ML) has changed the game in this industry by making it possible to create prediction models that are more flexible, accurate, and dynamic. These models can adapt to fit the energy needs of the future. These changes have made it much easier to optimise energy use in real time, manage loads that aren't linear, predict and keep an eye on loads, and guess and predict how much energy will be used in the short and long run. Machine learning is used in both short-term and long-term energy demand forecasting models to anticipate how much energy will be used over different time periods.

Machine learning techniques have been used a lot to make intelligent systems that can find and respond to changes in the context of smart buildings [125]. There are many distinct sorts of machine learning techniques, as shown by the Ugly Duckling and No Free Lunch theorems. The No Free Lunch theory says that 'no algorithm can be said to be better than any other.' If there is no background knowledge, any two algorithms could work just as well to solve the problem. The Ugly Duckling theorem says, 'We can't say that any two different patterns would be more similar to each other than any other pairs.' But this isn't true [126]. There are four main domains of machine learning, and each one deals with a different kind of learning job. There are different types of learning algorithms, such as supervised, unsupervised, semi-supervised, and reinforcement learning (RL). Figure 3 shows the Machine Learning models used in smart buildings for energy.

6.1 | Decision Tree

The decision tree method is a key way to use machine learning to make predictions. It builds a decision model from the real values of features in the data. Decision trees can be seen as a way to solve problems with classification and regression. In a tree structure, the leaves stand for class labels and the branches stand for the combinations of features that lead to those labels [127]. There is a regression tree when the target variable can take on continuous values. People generally say that decision trees are one of the best machine learning algorithms since they are fast and accurate. The most popular types of decision trees

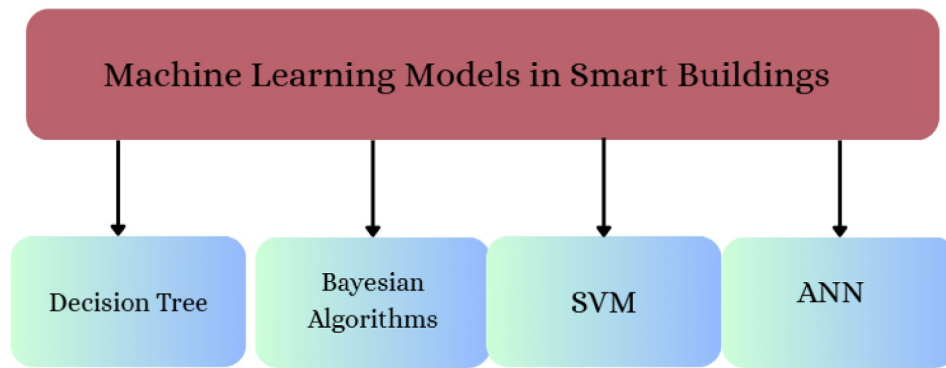


FIGURE 3 | Machine learning models used in smart buildings.

are conditional decision trees, chi-squared, M5, classification and regression trees, ID3, C4.5, and C5.0 [128].

The author of the study [129] presents a decision-tree-based machine learning methodology to rapidly simulate human behaviour in an intelligent environment. The purpose of the technique is to find the most prevalent ways that people act and how those actions are related to each other throughout time. The author of the study [130] offers a prototype of a distributed data mining system utilising the C4.5 classification algorithm for implementation in healthcare environments. This technology can be used to help with health services and keep an eye on patients. Decision trees provide a straightforward and comprehensible non-parametric methodology. One of the algorithm's biggest problems is that it might overfit.

6.2 | Bayesian Algorithms

Bayesian approaches use Bayes' theorem to solve difficulties with classification and regression. Naive Bayes, Gaussian naive Bayes, Bayesian belief network, and Bayesian network are the main types of Bayesian algorithms, as shown in reference [131]. The author of the study [51] proposes an indoor localization technique that employs Naive Bayes classification and dynamic temporal warping. They use the sensors included in cell phones to figure out which building the user is in and what they're doing there. The author of the study [132] devised a machine learning methodology employing Bayesian networks for the diagnosis of problems in HVAC systems. The model was created with the help of virtual sensors and a professional understanding of conservation laws, how different parts of a system work together, and past data. Both [133, 134] show how Naive Bayes classifier algorithms can be used to figure out what people are doing. The Naive Bayes approach can figure out what human activities are most likely linked to the sensor values that were seen.

6.3 | Support Vector Machines

Support vector machines (SVMs) are supervised machine learning techniques predominantly employed for classification tasks; they are also applicable to regression problems [135]. Support Vector Machines (SVMs) are widely employed to tackle many statistical learning problems, including facial and object recognition, text classification, spam detection, and handwriting analysis, among others [136]. The goal of SVM is to make the margin that separates the hyperplane from the nearest points of the two classes as big as possible. Support vectors are the points

that are on the edges, and the midpoint of the margin [137] is the hyperplane that best separates these edges.

The author of the study [138] developed an SVM technique to predict system-level electricity needs in public buildings that use energy sub-metrising systems. The author of the study [139] put forward a real-time human tracking system that uses support vector machines (SVM) to guess and find out where people are moving based on pictures taken by four cameras in smart homes. The author of the study [140] created a support vector machine (SVM) model that used just data from wireless motion sensors placed in each room to guess if guests were in the house. The author of the study [95] executed a study employing support vector machines (SVM) to autonomously recognise activities of daily living inside a smart home setting. The data was gathered from various sensors, such as door contacts, temperature and humidity sensors, microphones, infrared presence sensors, and others. The author of the study [141] developed a one-class classification method utilising one-class support vector machines for real-time activity fault detection in smart homes. The author of the study [142] developed a machine learning methodology employing support vector machines (SVM) and recurrent neural networks (RNN) to analyse temperature and heating source data for the purpose of identifying building occupancy behaviour within the context of energy efficiency consumption.

6.4 | Artificial Neural Network

An artificial neural network (ANN) is made to work like the networks in the brain. Using ANN models is a common way to solve difficulties with regression and classification [131]. Some common artificial neural network methods include the perceptron, backpropagation, Hopfield network, and radial basis function network (RBFN).

Artificial neural networks (ANNs) have a number of benefits, such as needing less statistical training, being able to find complicated nonlinear relationships between predictor factors and response variables, and being able to find all possible relationships between predictor variables [143]. On the other hand, its drawbacks include being a 'black box,' costing a lot of money to run, and being very likely to overfit. On the other hand, neural networks have some important limitations because of how they work. The most important ones are that they are hard to train without a local optimal solution, they can adapt to

changes in behaviour, they need to be validated, and it is hard to understand how well they work.

The author of the study [144] created a smart house system that uses pattern recognition based on artificial neural networks (ANNs) to make homes more energy-efficient. The technology has two parts: a recurrent neural network (RNN) that finds patterns in how people act and an artificial neural network (ANN) that is used for security in smart homes. Other researchers have used artificial neural networks to offer services that are aware of their surroundings. Campo et al. [145] devised a methodology that evaluates the probability of occupancy for each structural component and subsequently compares that probability with the prevailing conditions in a methodical manner. The author of the study [146] has a survey article that looks at how artificial neural networks can be used in smart home services. The author of the study [147] recommended employing a hybrid classifier technique to recognise activities such as rowing, riding, playing football, walking, running, sitting, and hiking. This approach would utilise a tree structure that incorporates a priori information and artificial neural networks. The author of the study [148] introduced a home energy management system idea that employs a neural network algorithm to forecast the electricity generation of the photovoltaic plant (PV) and the household energy consumption within a certain timeframe.

6.5 | Unsupervised Learning

Unsupervised learning is the process of building algorithms that use data without labels. It is used to look at the behaviour or system being studied [149]. As a result, the algorithm doesn't know what factors led to the outcome. This means that the unsupervised learning algorithm puts the input samples into different groups based on how similar they are to each other. The clustering method uses a number of different characteristics taken from the data to find relationships that aren't immediately clear. During model preparation, the general rules of the model are found and prepared by using the inferred structures that exist in the input data. A mathematical method that sorts data by how similar it is or that always reduces redundancy could do this [150].

Using the unsupervised strategy, which is helpful when labels for the data utilised are not easily available [151], some smart building characteristics have been found. People often have trouble with grouping, dimensionality reduction, and learning association rules when they learn unsupervised. There are many different types of unsupervised learning methods, and they are used a lot. The Apriori algorithm and k-Means are two examples of supervised learning algorithms. Unsupervised learning typically lacks an output metric; its objective is to identify patterns and correlations within a dataset, requiring only the capacity to discern features [152].

Unsupervised learning is most notable for not having a clear way to learn. So, the learner might not get any meaningful information from the set of traits that were picked for training. The purpose of clustering, which is a sort of unsupervised learning, is to detect patterns in the data by putting data points that are similar in theme into smaller groups called K-clusters. You can find patterns in the data with this method. In the next sections, we will look at clustering problems and association

problems, which are two types of unsupervised learning tasks. Table 7 shows the ML Learning Techniques in Smart Buildings.

7 | Discussion

7.1 | Stages of ML Implementation in Building Energy Management Systems

As indicated in Figure 3, using machine learning in energy management systems usually goes through three steps:

The major goal of this first phase is to figure out if the programme is technically possible and if it is strategically necessary to use machine learning to improve the energy efficiency of buildings. To do this, you need to first figure out what problems machine learning can address well, and then choose the appropriate data formats and algorithms for the job. To make plans that really protect energy resources, model designers need to be able to combine their computer science skills with a deep understanding of the rules of energy management.

It is important to figure out the right amount and type of data that will be needed, the exact machine learning method that will be used, and finally, if the whole modeling project will really make the organisation more energy efficient during the planning phase [10, 12].

During the testing phase, we will observe how well ML technology predicts results and how well the proposed algorithms perform with building energy data. In this process, models are created, and the data is meticulously prepared. Post-collection data processing is carried out to enhance the quality of the data and ensure that there is sufficient data for robust ML model construction. Although there are methods for gathering and processing data, there are no reliable quantitative tests to determine whether the data is sufficient. This leads to different research projects having different amounts of data, which is a major problem at this point [10, 12].

Adding a verified machine learning model to an existing building's energy management system is the penultimate step in the application process. Improving system performance and decision-making are the main objectives.

This integration can be done in a number of ways. For example, it can entail swapping out the existing prediction model for a state-of-the-art machine learning model. This would essentially replace the previous approach, but it would require new methods for gathering and handling data. Another option would be to enhance and modify the current system to enable it to incorporate fresh data patterns discovered by the machine learning models to provide more management information and reasoning for decisions. The majority of current research [10] focuses on improving prediction models in existing systems. Figure 4 shows the stages of ML implementation in energy management systems.

7.2 | Case Studies

7.2.1 | Real-World Applications of AI in Smart Buildings

Machine learning has changed the way smart buildings are managed and run, making them far better at saving energy, being

TABLE 7 | Summary of ML learning techniques in smart building.

Ref.	Year	Method	Key findings
[153]	2022	Machine learning	This study carried out a thorough analysis of artificial intelligence use in district communities that are carbon neutral. The assessment examined district demand, energy storage, energy supply, and energy management. There are categories and underlying mechanisms that have been identified by several machine learning techniques, including deep learning, reinforcement learning, supervised learning, and unsupervised learning. Next, we will discuss some applications of machine learning, including advanced energy management, district energy demand, hybrid energy storage, and renewable energy.
[154]	2024	ML, demand prediction	This study examined the application of machine learning algorithms for energy management forecasting in modern buildings (MBs). Hybrid and ensemble machine learning techniques, such as support vector machines (SVMs) and random forests (RF), outperform single prediction models. Hybrid machine learning models may be able to predict energy use with up to 15% greater accuracy than single machine learning models. The results show that machine learning methods can be utilised to control energy in MBs in a precise and useful way. Adding new parameters to the information could also make energy prediction systems more accurate and useful.
[155]	2020	Machine learning (SVM, decision trees), deep learning	This is done by examining diverse machine learning tools, frameworks, and applications, and subsequently organising the taxonomy according to these criteria. In conclusion, the paper gives a short overview of the works that were given, emphasising on the problems, unanswered questions, and possible paths for future research in this area.
[17]	2020	Machine learning	This study examines research regarding the utilisation of machine learning (ML) models in smart buildings to forecast occupancy and window-opening behaviour. The indoor concentration is an important sign that can help it guess which individuals will be in a room. Use feature engineering to make predictor variables that are more relevant to improve the model's performance even more.
[156]	2021	Machine learning, wireless sensor networks	The researchers promote the adoption of machine learning techniques as a means to enhance the efficiency of traditional WSN-IoT nodes employed in smart city projects. To the best of the author's knowledge, this is the first comprehensive review of all machine learning methods used in low-power wireless sensor networks (WSN-IoT) for smart cities. This unique survey report shows that supervised learning algorithms are used the most in smart city applications (61%), followed by reinforcement learning (27%) and unsupervised learning (12%). Unsupervised learning is only used one percent of the time.
[121]	2025	Machine learning	This article gives a full look at the machine learning methods used for finding faults, predicting energy use, scheduling energy use, and controlling operations. The research and literature review indicate that supervised learning methodologies, such as support vector machines and random forests, exhibit significant classification accuracy for defect detection; nonetheless, they necessitate extensive, specially annotated datasets. Unsupervised learning methods, such as principal component analysis and clustering algorithms, provide efficient defect detection capabilities in the absence of labelled data. Still, these methods might have trouble with complicated nonlinear patterns.

(Continues)

TABLE 7 | (Continued)

Ref.	Year	Method	Key findings
[157]	2025	Machine learning	This research examines the application of machine learning (ML) in sustainable design (SBD), intelligent energy management (IEM), and building performance predictions. It shows how intelligent building systems use real-time data analysis and forecasts to make energy consumption more efficient and lower carbon emissions. This paper not only gives a short history of machine learning (ML) and its prospects, but it also talks about the big problems that the construction sector has to deal with. The results provide a technology-driven perspective on enhancing sustainability practices within the construction industry and underscore intriguing directions for further research.
[100]	2025	Machine learning	Long-short-term memory has a significant impact on predictive modelling, as evidenced by the R-squared value of 0.993 and the mean squared error of 0.004. With R-squared values of 0.462% and 0.9%, respectively, support vector regression and XGBoost demonstrated that they were only capable of making limited predictions. Long-short-term memory is the best model for predicting energy performance, according to this study, giving engineers and architects the information they need to choose energy-efficient building designs.
[101]	2025	Machine learning, deep learning, RL	This study seeks to provide a comprehensive analysis of current research trends in machine learning-based energy systems, clarify the related issues, and suggest prospective research directions and recommendations. This study provides a thorough examination of machine learning-driven approaches across several domains of energy systems, including energy policy and sustainability, in contrast to previous studies that focus solely on machine learning applications within the renewable energy sector. The goal of this project is to create a complete resource that links academic progress with real-world uses in energy systems, using new ideas based on machine learning.

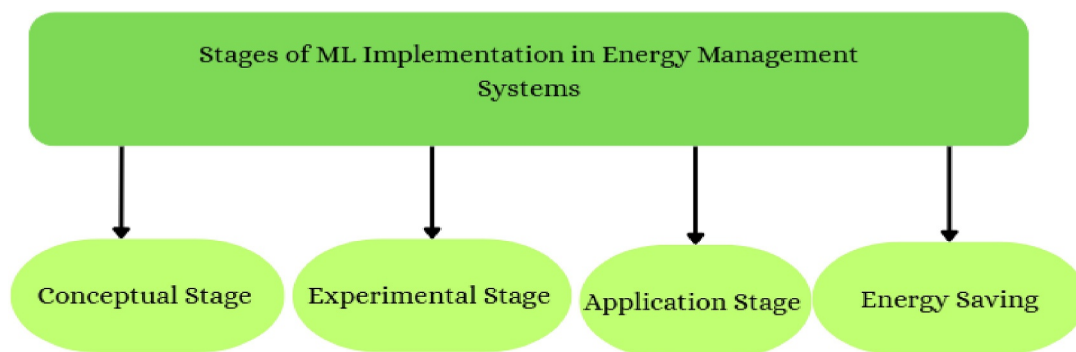


FIGURE 4 | The stages of ML implementation in energy management systems.

reliable, and making people feel comfortable [35]. It shows the machine learning applications that have worked well in smart buildings. Table 8 shows the summary of case studies.

7.2.2 | Comparative Evaluation of ML Implementations in Smart Buildings

Before choosing a predictive method for smart building management, think carefully about the goals, data elements, and application limits. Support Vector Machines, decision trees, and random forests are better for modelling sophisticated, non-

linear interactions, although linear regression and time series analysis are better for basic forecasting and trend analysis. Deep learning models like neural networks are great for looking at data with a lot of dimensions, but they also need a lot of resources and data. Clustering makes predictive modelling more accurate by putting related data sets together.

Table 8 provides a comprehensive analysis of the many practical implementations of machine learning in intelligent buildings. This comparative review examines a range of criteria to find context-specific elements that affect machine learning performance.

These include measurable results, algorithmic methods, sensor infrastructure, geographic location, climate zone, and building type. A variety of architectural contexts demonstrate variations in the installation and efficacy of ML systems, as illustrated in the Table 9. For example, Google's DeepMind project in data centres uses a lot of operational data that is collected very often.

8 | Challenges and Opportunities

8.1 | Diverse Research Objects

Although the data mining capabilities of ML models can be effectively showcased using diverse research objects in building energy prediction challenges, it is essential to assess the prospective benefits of using heterogeneous research objects for the application of ML models in real-world scenarios. There has been research on energy prediction, but it is still not clear which problems are the most important and best suited for machine learning solutions. The importance of time-scale and predictive factors in improving building energy performance remains disputed. But the concept underlying ML's rapid growth and widespread use in many fields is that it can lead to groundbreaking discoveries. Before the model

can adapt to the data in that domain, identifying viable application opportunities is crucial. Because work items aren't always the same, it's harder to compare and choose ML algorithms and non-standardised data formats. To address this issue, it is essential to integrate market demand analysis with a technological feasibility assessment to pinpoint the key building energy efficiency challenges that remain unresolved through ML technology.

8.2 | Diverse ML Algorithms

The quick development of machine learning has led to the use of many algorithms in studies to improve building energy efficiency. Numerous algorithms have been employed over time. Support vector machines (SVMs), multiple linear regression (MLR) [158], extreme learning machines (ELMs) [159], deep learning (DL) [160, 161], decision trees (DTs) [162] random forests (RFs) [163] and artificial neural networks (ANNs) [159] are some of the methods. More than 128 machine learning techniques have been applied in various investigations, according to a recent study [164]. Nevertheless, no algorithm has been acknowledged as being totally infallible [3]. No single machine learning technique is superior to all others in every

TABLE 8 | Details of case studies.

Case study/Location	Building type	AI/Deep learning technique	Description of implementation
Google DeepMind data centres (USA)	Data center	Deep reinforcement learning (DRL)	DeepMind AI-controlled cooling systems use real-time sensor data to optimise HVAC operations.
The edge building, Amsterdam (Netherlands)	Office building	Machine learning, IoT analytics	AI analyzes employee movement and energy usage to optimise lighting and HVAC systems.
Empire state building retrofit (USA)	Commercial skyscraper	Predictive analytics, ML	AI-based analytics are used to predict energy demand and optimise building operations.
Singapore smart buildings programme (Singapore)	Smart city buildings	Deep learning, computer vision	AI models analyse camera and sensor data to monitor occupancy and optimise space utilisation.
Siemens smart infrastructure projects (Germany)	Industrial & commercial buildings	AI & digital twins	AI-powered digital twin models monitor building systems and predict equipment failures.
IBM Watson smart buildings (Global)	Corporate buildings	NLP, ML, deep learning	Watson AI integrates data from multiple building systems for intelligent decision-making.

TABLE 9 | Comprehensive analysis of the many practical implementations of machine learning in intelligent buildings.

Entity	Building typology	Approach	Outcome
Google DeepMind	Industrial	Optimisation of cooling using deep RL	Reduction of cooling energy by up to 40%
Stanford Univ	Campus/mixed-use	The use of regression-based machine learning for anticipating HVAC demand	The entire campus's energy savings of 10%
The edge (deloitte)	Smart building	Decision trees and real-time sensor data for air conditioning, heating, ventilation, and lighting	70% energy consumption reduction
Copenhagen Smart city	Commercial	Using supervised machine learning for maintenance that is based on predictions	The path toward carbon neutrality that has been made possible
Carnegie mellon University	Mixed-use city-scale infrastructure	Using IoT and machine learning to predict demand in cities	A roadmap for achieving carbon neutrality through the use of
NTU EcoCampus	Research campus	A mix of artificial neural networks and support vector machines for predicting energy use at the appliance level	Energy savings of 28% for the building:

TABLE 10 | Major data types used in ML-based building energy modeling.

Data type	Examples
Information about meteorology	Climate data includes things like sunshine intensity, wind speed and direction, humidity, and rainfall
Data about the indoor environment	The interior temperature, lighting conditions, and air quality (including CO ₂ and volatile organic compounds) of a structure
Occupancy-related data	A building's occupancy describes how many people reside there, what they do there, and how they act
Time index	The ability to look at and connect data throughout time using a timestamp or time index that is related to the data that was collected
Properties of building	This category includes the following parameters: Orientation, surface area, wall area, roof area, overall height, glazing area, shading coefficient, building envelope heat transfer coefficient, and window-wall ratio
Past data	Energy use patterns, electricity bills, and weather records from the past

circumstance, even when multiple approaches have been developed to address the same issue [10]. Due to the diversity of the experimental variables used, the results of studies using real-world construction situations cannot be generalised. The results vary as a result of this. Experimental designs may become inconsistent due to variations in data types, data amount, and prediction outputs. The evaluation criteria, including processing time, mean absolute error (MAE), and coefficient of variation (CV), are all distinct, which exacerbates the lack of an efficient approach. The focus on using a variety of evaluation criteria to assess the efficacy of machine learning systems limits research. Because every building is unique and people have various preferences, the application of machine learning to increase building energy efficiency has slowed slowly. The best algorithms for resolving building energy prediction issues can be found by exchanging data and establishing evaluation standards. This increases the field's overall efficiency and facilitates the comparison of various projects.

8.3 | Unstructured Data

The current research faces a substantial practical obstacle stemming from the absence of a uniform data structure, which is crucial for the progress of machine learning-driven building energy modeling. The data used to create energy models included [165], has been aggregated and categorised (see Table 9 for further details). When choosing input variables and sample rates, researchers generally look at the properties of the prediction objects and the availability of data. As a result, the amount and type of data used in different studies are not the same. A significant problem is that the basic problems of creating energy management for machine learning model analysis are still not clear.

The numerous prediction objects make it harder to combine the data structure that goes with the prediction elements [10]. The construction of a universal standard for data collection is hindered by disparities in data structures, hence complicating the examination of similarities and differences across various study outcomes. Models that use machine learning to make energy predictions are not very common [166]. It is very important to figure out what the main problems that machine learning can solve are and to summarize the relevant data sources. To

overcome the unique prediction problem, the data structure needs to be changed.

8.4 | Technology-Oriented

To make machine learning applications work, it's important to find problems that have a lot of potential for use. These problems should be solvable by machine learning models, and the answers should have a substantial or revolutionary effect on the domains where they are utilised [9]. Researchers are urged to concentrate on assessing the capacity of machine learning models to learn from building energy data within the technology-oriented research framework [12]. There has been little research regarding the efficacy of the proposed machine learning models for building energy management systems and the feasibility of the prediction challenges. This is the primary reason why earlier studies have exhibited inconsistent research subjects and data formats. Because of this, there will be differences between what practitioners need and what current research shows. Researchers should follow a problem-driven research methodology that looks at all the processes involved in using machine learning models in order to effectively promote the use of this technology to create energy-saving strategies. Table 10 shows the data types with examples.

9 | Conclusion

This paper provides a comprehensive overview of prior studies on forecasting energy consumption in smart buildings using AI, Deep learning and Machine learning methods. Traditional approaches, while easier to implement, often lack the accuracy needed to account for complex factors such as weather and indoor conditions. AI techniques, on the other hand, excel at handling non-linear relationships and offer superior forecasting performance, demonstrating adaptability across various research contexts. This paper presents an overview and several examples of energy consumption projections in intelligent buildings.

It assesses research on various building forecasting methods for residential and nonresidential buildings, including physical, data-driven, and hybrid approaches. The various forecasting techniques, the input components, and the prediction horizon used for each technique type were examined by the researchers. It was

determined whether the measures employed to validate the procedures were accurate. The article's conclusion discusses the conclusions drawn from each of the study methodologies and issues that were looked at. Using data-based and hybrid approaches that prioritise energy for the entire building, much of the study was done on nonresidential structures, allowing for consideration of all existing systems in the buildings.

Author Contributions

Sundus Munir: resources, validation, writing – original draft. **Muhammad Tayyab Khan:** resources, validation, visualization. **Tehseen Mazhar:** resources, software, supervision. **Tariq Shahzad:** data curation, formal analysis, validation, visualization. **Muhammad Adnan Khan:** software, supervision, writing – original draft, writing – review and editing. **Mamoon M. Saeed:** data curation, formal analysis, resources, validation. **Habib Hamam:** methodology, project administration, writing – original draft, writing – review and editing.

Funding

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding authors upon request.

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